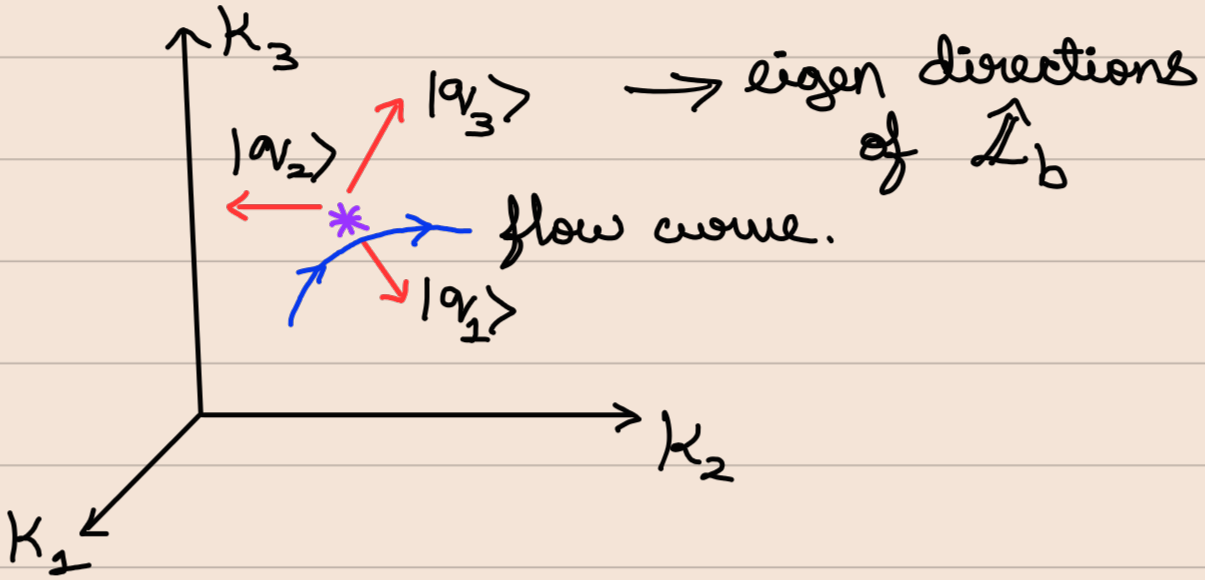
02/04/25

Lecture 24

RG flow curves & Universality classes

Under the action of RG, many times near a fixed point in the space of coupling constants $\{K\}$, this results in a flow determined by the variations $\{h_m\}$ along the eigen directions of $\hat{\mathcal{L}}_b$.



To get the flow curve equations, we use the fact that, $h_m^{(\ell)} = h_m^{(0)} b^{\ell \lambda_m}$

$$\rightarrow \left[\frac{h_m^{(\ell)}}{h_m^{(0)}}\right]^{1/\lambda_m} = b^{\ell} = \text{constant}$$

$$\Rightarrow \left[\frac{h_1^{(\ell)}}{h_1^{(0)}}\right]^{1/\lambda_1} = \left[\frac{h_2^{(\ell)}}{h_2^{(0)}}\right]^{1/\lambda_2} = \left[\frac{h_3^{(\ell)}}{h_3^{(0)}}\right]^{1/\lambda_3} = \cdots\cdots = b^{\ell}$$

Thus we have a parametric eq^n for $\{h_m^{(\ell)}\}$ with 'ℓ' as a parameter.

Since, $h_m^{(\ell)} = h_m^{(0)} b^{\ell\lambda_m}$, based on the sign of λ_m.

- If $\lambda_m > 0 \Rightarrow h_m^{(\ell)}$ grows with 'ℓ'. Then h_m is a relevant variable.

- If $\lambda_m < 0 \Rightarrow h_m^{(\ell)}$ decreases with 'ℓ'. Then h_m is an irrelevant variable.

- If $\lambda_m = 0 \Rightarrow h_m^{(\ell)}$ remains constant with respect to 'ℓ'. Thus its relevance is determined by high order analysis.

RG for two Couplings

Let $\vec{K} = \{K_1, K_2\}$ be 2 coupling constants defining some hamiltonian. Let K_1^* & K_2^* be fixed points. Then in the basis of $\hat{\mathcal{L}}_b$ the $\Delta\vec{K}$ can be written as, $(h_1^{(0)}, h_2^{(0)})$.

$$\Rightarrow \hat{\mathcal{L}}_b^{\ell}(h_1^{(0)}, h_2^{(0)}) = \underbrace{\left[h_1^{(0)} b^{\ell\lambda_1}, h_2^{(0)} b^{\ell\lambda_2}\right]}$$

Can be plotted parametrically by using b^{ℓ} as a parameter.

Or we can eliminate the parameter as follows.

$$h_1^{(\ell)} = h_1^{(0)} b^{\ell\lambda_1}$$

$$h_2^{(\ell)} = h_2^{(0)} b^{\ell\lambda_2}$$

$$\Rightarrow \left[\frac{h_1^{(\ell)}}{h_1^{(0)}}\right]^{\frac{1}{\lambda_1}} = \left[\frac{h_2^{(\ell)}}{h_2^{(0)}}\right]^{\frac{1}{\lambda_2}} = b^{\ell}$$

$$\rightarrow h_2^{(\ell)} = h_2^{(0)} \left[\frac{h_1^{(\ell)}}{h_1^{(0)}}\right]^{\frac{\lambda_2}{\lambda_1}}$$

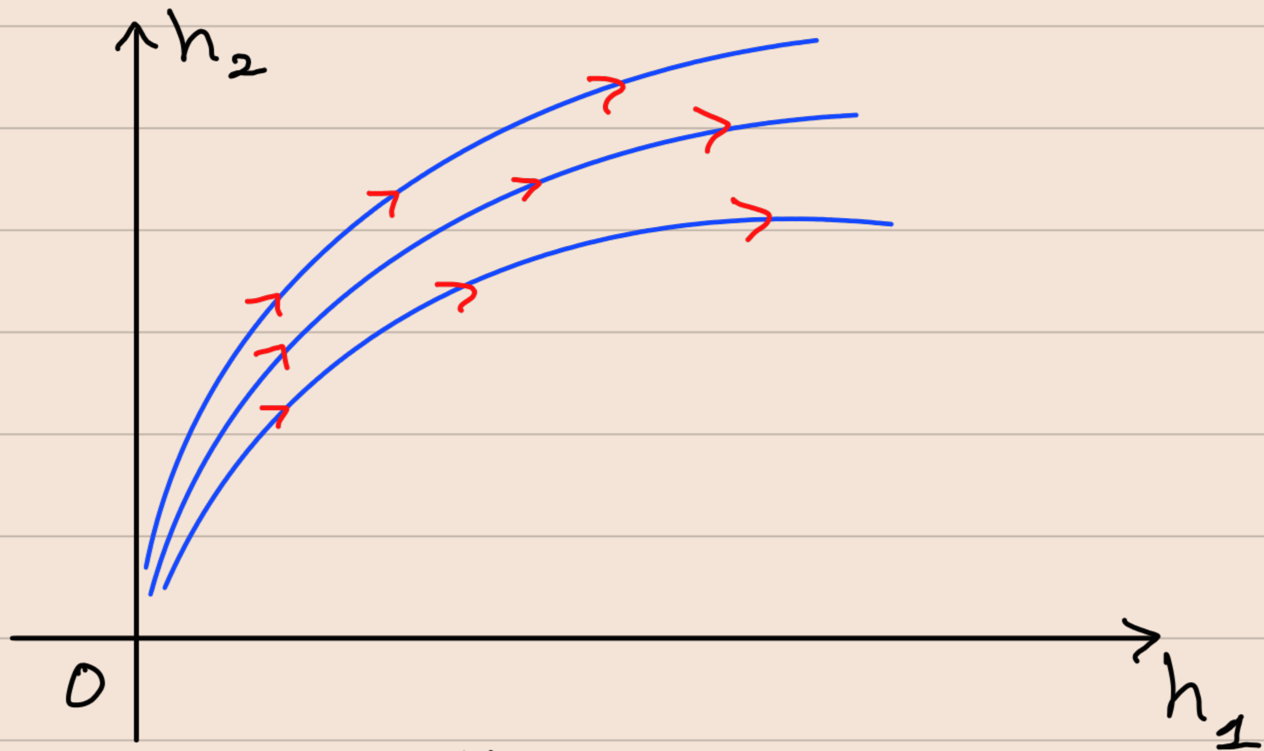
• <u>If $\lambda_1 > \lambda_2 > 0$</u> $\Rightarrow h_1^{(\ell)} = h_1^{(0)} b^{\ell\lambda_1}$

$$h_2^{(\ell)} = h_2^{(0)} b^{\ell\lambda_2}$$

$$\rightarrow h_2^{(\ell)} = h_2^{(0)} \left[\frac{h_1^{(\ell)}}{h_1^{(0)}}\right]^{\frac{\lambda_2}{\lambda_1}} = \text{Constant} \times \left[h_1^{(\ell)}\right]^{\phi}$$

$$\phi = \frac{\lambda_2}{\lambda_1} < 1$$

$$\Rightarrow \underline{h_2^{(\ell)} \propto \left[h_1^{(\ell)}\right]^{\phi} \quad (\phi < 1)}$$



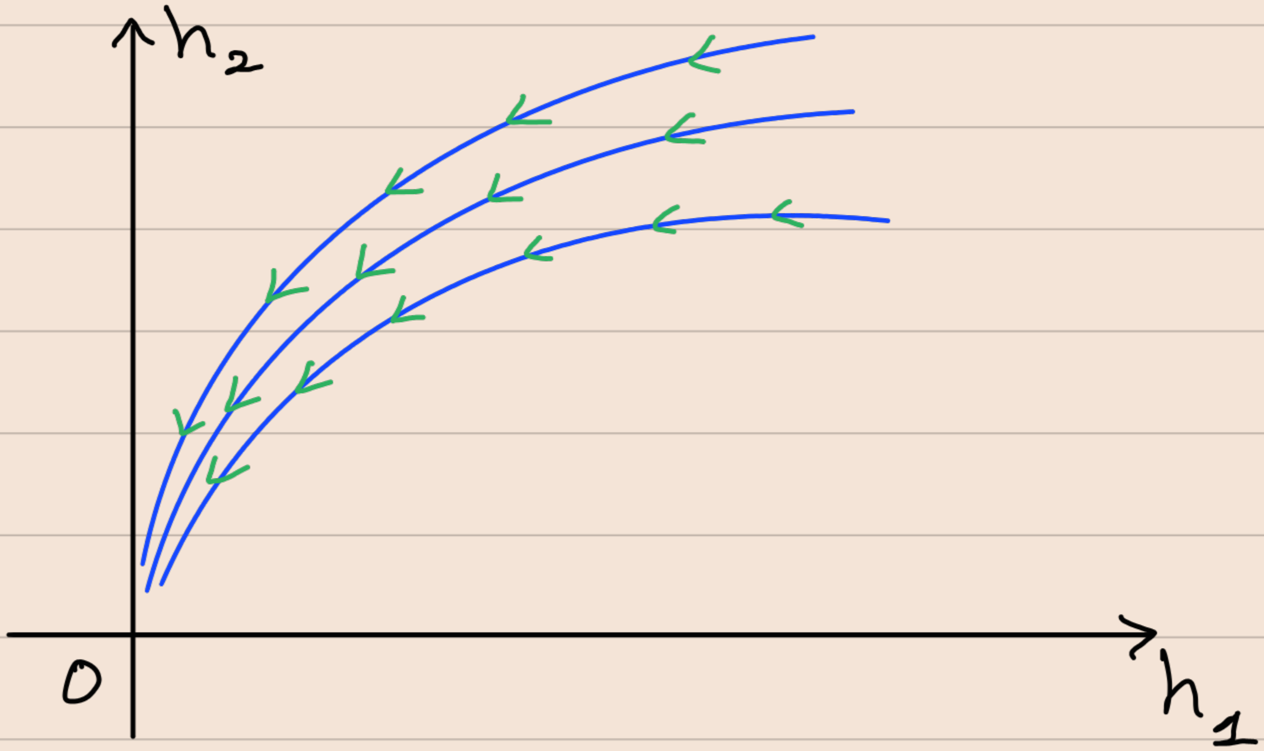
The values of $h_1^{(\ell)}$ & $h_2^{(\ell)}$ increases with 'ℓ'. Hence the fixed point is unstable. All flow curves with different initial conditions will eventually go away from the fixed point.

• <u>If $\lambda_1 < \lambda_2 < 0$</u>, <u>$|\lambda_1| > |\lambda_2|$</u>

$$\Longrightarrow \quad h_1^{(\ell)} = h_1^{(0)}\, b^{-\ell|\lambda_1|}$$
$$h_2^{(\ell)} = h_2^{(0)}\, b^{-\ell|\lambda_2|}$$

$h_1^{(\ell)}$ & $h_2^{(\ell)}$ will decrease with 'ℓ'

$$\Longrightarrow \quad h_2^{(\ell)} = h_2^{(0)} \left[\frac{h_1^{(\ell)}}{h_1^{(0)}}\right]^{\frac{|\lambda_2|}{|\lambda_1|}}$$



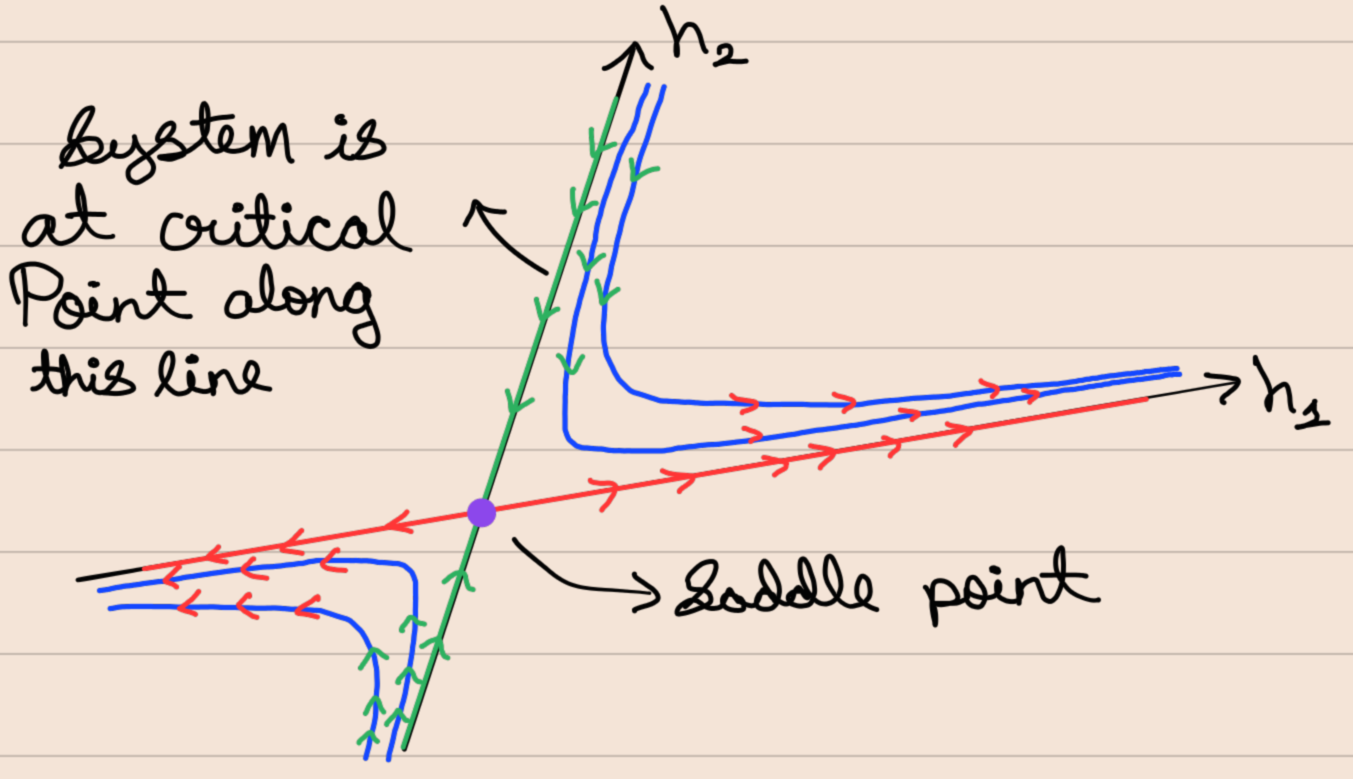
The values of $h_1^{(\ell)}$ & $h_2^{(\ell)}$ decreases with 'ℓ'. Hence the fixed point is stable. All flow curves with different initial conditions will eventually go towards the fixed point.

• <u>If $\lambda_2 < 0 < \lambda_1$</u>, <u>$|\lambda_1| > |\lambda_2|$</u>

$$\Rightarrow \left.\begin{aligned} h_1^{(\ell)} &= h_1^{(0)}\, b^{\ell\lambda_1} \\ h_2^{(\ell)} &= h_2^{(0)}\, b^{-\ell|\lambda_2|} \end{aligned}\right\} \Rightarrow h_2^{(\ell)} = h_2^{(0)} \left[\frac{h_1^{(\ell)}}{h_1^{(0)}}\right]^{-\frac{|\lambda_2|}{\lambda_1}}$$

$$\Rightarrow h_2^{(\ell)} \left[h_1^{(\ell)}\right]^{\frac{|\lambda_2|}{\lambda_1}} = \text{constant}$$

Here the line $h_2 = 0$ is unstable i.e h_1 axis is unstable as $\lambda_1 > 0$
And the line $h_1 = 0$ is stable, i.e., h_2 axis is stable as $\lambda_2 < 0$.
Thus here the fixed point is a saddle.



Unstable and Stable Manifold

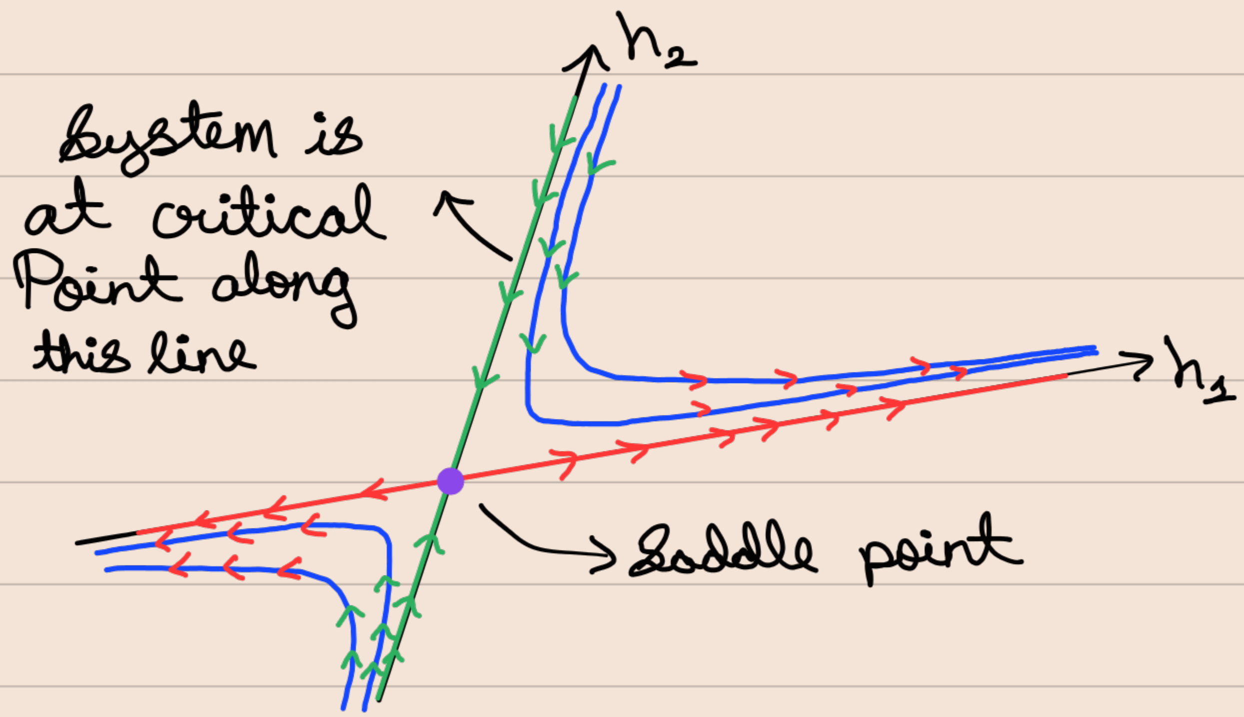
Unstable manifold near $\vec{K}^*$ fixed point is the set of all couplings which flow away from the fixed point under the RG flow. Near the fixed point, this manifold is tangent to the space spanned by the relevant eigen vectors whose $\lambda_m > 0$.

Thus to reach the fixed point we need to tune the system such that under repeated application of RG, system reaches criticality. This is possible by setting all relevant variables to zero initially.

Then the tuned $\Delta\vec{K}_{tuned}$ will lie on a stable manifold which is the set of all points that flow into the fixed point and is tangent to the space spanned by irrelevent eigen vectors $(\lambda_m < 0)$.

The dimensions of unstable manifold is equal to the number of revelant variables & the dimensions of the stable manifold is equal to the number of irrelevant variables.

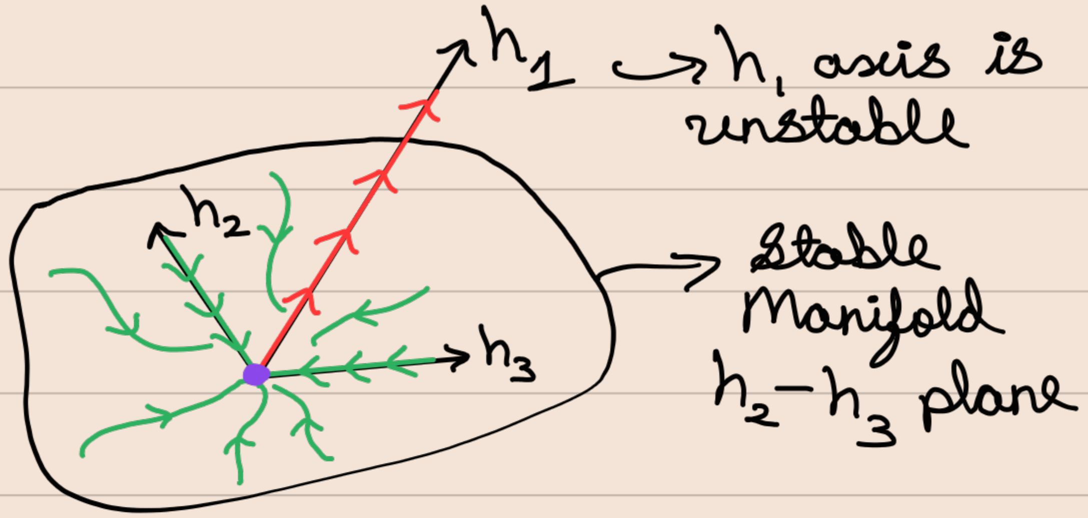
In the below example, the line $h_1 = 0$, where



- $h_1 \rightarrow$ relevant direction, hence h_1-axis is unstable manifold
- $h_2 \rightarrow$ irrelevant direction, hence h_2-axis is stable manifold.

The Systems at the h_2-axis $(h_1 = 0)$ are all at the critical behaviour.

Eg. In a 3 coupling (K_1, K_2, K_3) Space we can have a fixed point $\vec{K}^*$ such that h_1 is relevant $(y_1 > 0)$ & h_2, h_3 are irrelevant $(y_2 < 0$ & $y_3 < 0)$.



Those systems which lie in Stable manifold will eventually reach the fixed point at the origin of (h_1, h_2, h_3)

Universality Class: Thus all those systems which are in the stable manifold will be at or near criticality & will exhibit the universal critical behaviour governed by that fixed point, with same critical exponents & scaling laws. Thus every saddle fixed points will define its own universality class.